Certificate of Internship

This is to certify that Cao Yu (ID No.: 320511200508084514), a student from Xi'an Jiaotong-Liverpool University, served as a Vision Algorithm Intern at Suzhou HYC Technology Co., Ltd. for a four-week internship from 9 October 2026 to 5 November 2026.

During the internship, the student was attached to the R&D Department and worked on display-panel defect inspection and controllable defect simulation for machine-vision applications. The main responsibilities were as follows:

1. **Literature and problem definition.** Reviewed public literature, public datasets and open-source implementations on display-panel defect inspection; studied defect categories, imaging and inspection workflows, unsupervised anomaly detection for low-contrast Mura and line-like defects, and the applicability of public datasets; and compiled a structured survey of 22 sections that served as the basis for topic selection and experimental planning.

2. **Controllable defect simulation.** Implemented a parametric simulator that controls the position, extent, shape, size, orientation, polarity and intensity of Mura-like and line-like appearance changes on normal backgrounds; established a verifiable labelling scheme that separates the requested region, the continuous perturbation field, the actual pixel-level residual and the effective label region; and validated the simulator on 24 pre-defined control requests and a 480-request shared hold-out set, of which 460 were valid and 20 invalid, with every invalid case and its cause retained.

3. **Evaluation on real industrial data.** Built an evaluation pipeline covering native-resolution windowed inference, exclusion of padded borders, exact global pixel-wise average precision, image-level AUROC and fixed-threshold alarm counting; and carried out two rounds of paired training on KolektorSDD2 under a fixed five-real-anomaly setting with three initialisations in each round, reporting a small positive but unstable effect of synthetic substitution together with its false-alarm and missed-detection costs.

4. **Detector response measurement and statistical analysis.** Measured the response of a frozen PatchCore detector to 7,560 digital stimuli built from 60 backgrounds, 9 geometries, 7 intensity levels and 2 polarities, and quantified both the alarm rate and the spatially-joint alarm rate together with the resulting dose-response trend; fitted six candidate statistical representations under a pre-registered protocol; and completed an independent audit of earlier measurement definitions, which led to the revision of four previously overstated conclusions.

5. **Engineering practice and reporting.** Worked with isolated Python environments, frozen configurations and read-only verification entries so that every reported number remains traceable and reproducible; kept all negative and failed results without re-selection; and prepared stage reports and presentation materials summarising the work and its limitations. All model training runs were executed manually under the supervisor's guidance.

During the internship, the student was diligent and responsible, completed the assigned research tasks on schedule, and demonstrated good learning ability, experimental discipline and independent judgement, in particular the willingness to verify measurement definitions and to revise earlier conclusions when the evidence did not support them. Through this project, the student became familiar with the complete workflow of a vision-inspection research task, from literature review and experimental design to data curation, model evaluation and technical reporting, and gained a clear understanding of what a controlled experiment on industrial defect data can and cannot support.

This certificate is hereby issued.

Internship Unit: Suzhou HYC Technology Co., Ltd.

Department: R&D Department

Reference Person:

Contact Number:

Date: 9th November 2026